

Applied A.I. Solutions

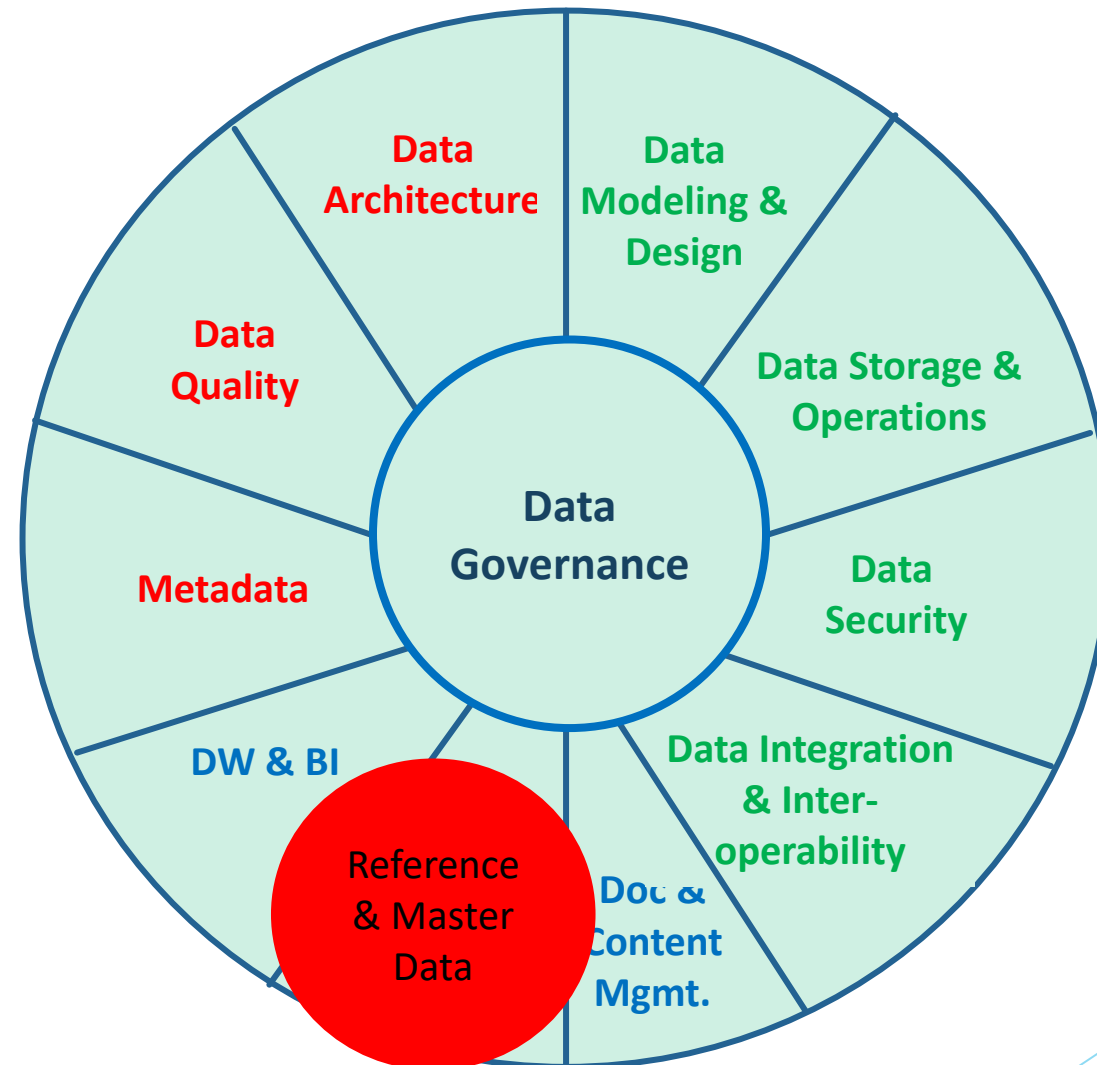
Foundations of Data Management

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REFERENCE AND MASTER DATA MANAGEMENT

The DAMA Wheel



INTRODUCTION

- In any organization certain data is required **across** business areas, processes, and systems
- Everyone **benefits** if they can access to the **same** customer lists, geographic location codes, accounting costs centres, and other data used to run the business
- **Reference and Master Data** are used to avoid inconsistencies in data structures and data values between systems

R&MD Management Framework

Definition

Managing shared data to meet organizational goals, reduce risks associated with data redundancy, ensure higher quality, and reduce costs of data integration

R&MD programs are guided by the following principles:

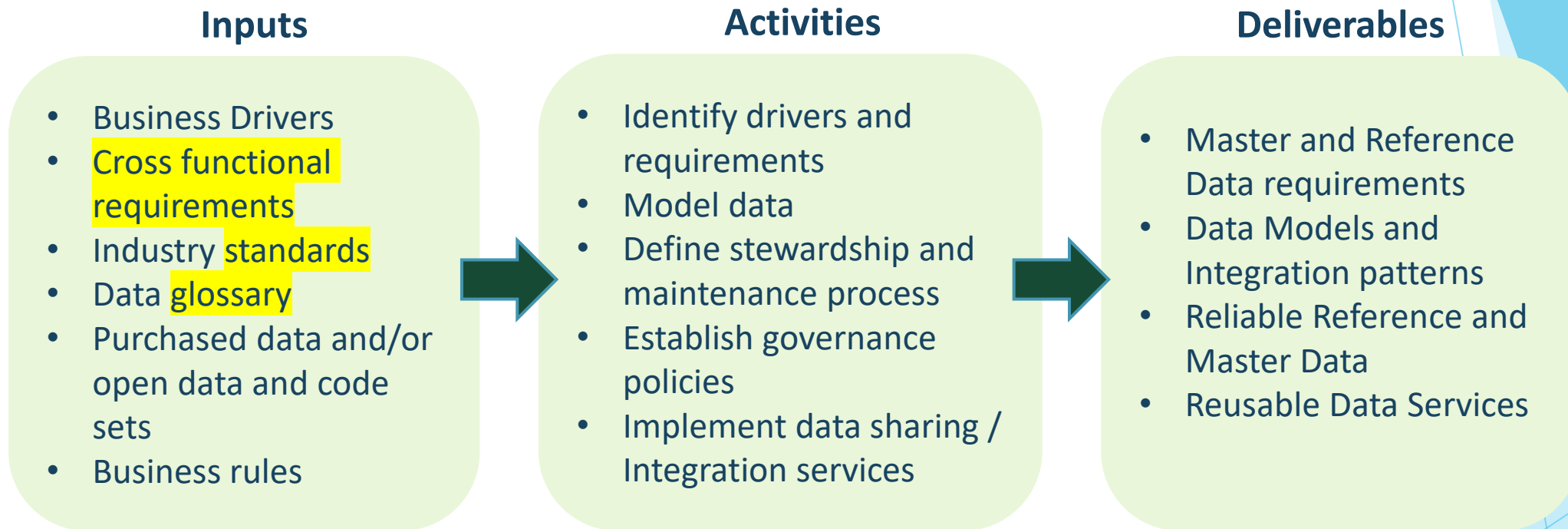
- Shared Data
- Ownership
- Quality
- Stewardship
- Controlled Change
- Authority

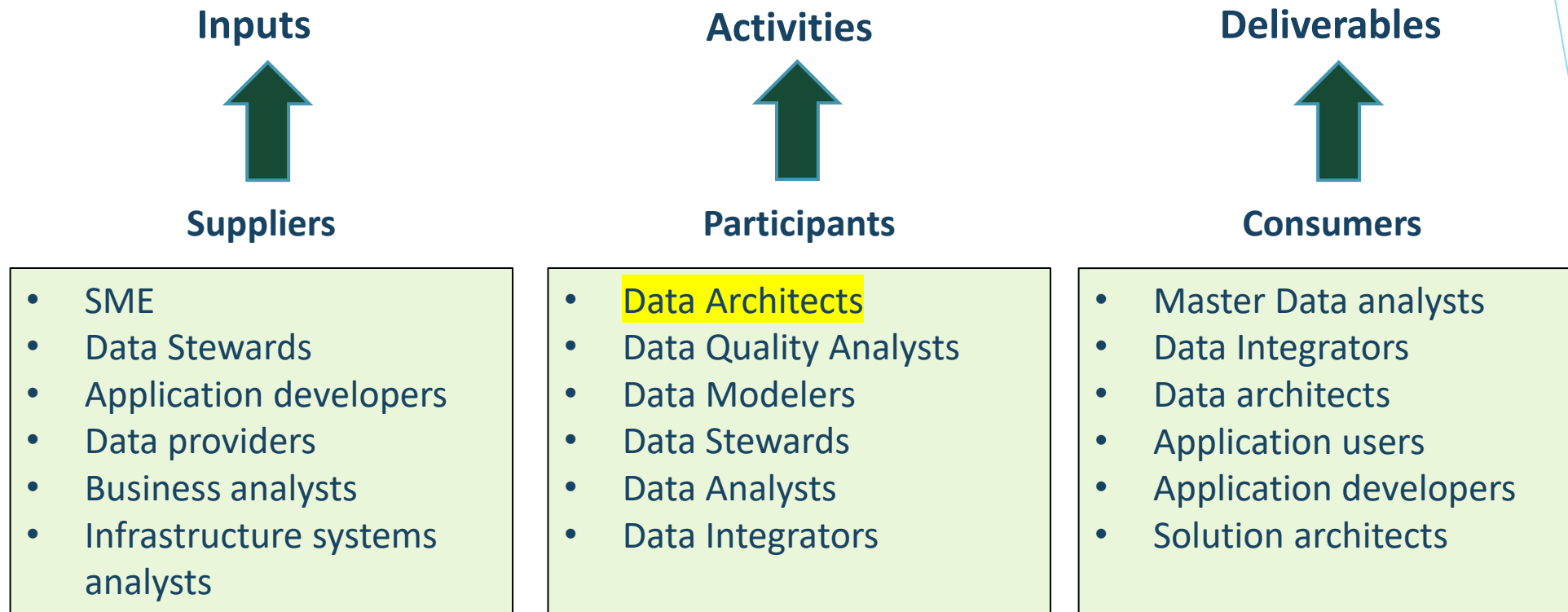
Goals

1. Enable sharing of information assets across business domains and applications
2. Provide authoritative source of reconciled and quality-assessed master and reference data
3. Lower cost and complexity through use of standards, common data models, and integration patterns



Business Drivers







Technical Drivers

Techniques

- Conditions-of-use agreements
- Business key cross references
- Processing log analysis

Tools

- Data Modeling tools
- Metadata repositories
- Data profiling and quality tools
- MDM application platforms
- Data sharing / Integration architecture

Metrics

- DQ and compliance
- Data Change activity
- Data Consumption, services
- Data sharing availability
- Data Steward coverage
- Data sharing volume, utilization

Master Data

- It is “the **data that provides the context for business activity** in the form of abstract concepts. It includes definitions and identifiers, and external objects involved in business transactions, such as **customers, products, employees, vendors, and other controlled domains** (code values)”

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² Malcom Chisholm proposed a six-layer taxonomy of data that includes Metadata, Reference Data, enterprise structure data, transaction structure data, transaction activity data, and transaction audit data (Chisholm, 2008; Talburt and Zhou, 2015)

It represents business objects which contain the **most valuable**, agreed upon information **shared** across an organization

Master Data are considered ...

- **Systems of Record:** where data is created or captured, and/or maintained through a defined set of business rules
- **Trusted Source:** it is recognized as the “best version of the truth”

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Master Data Management

A discipline in which **business and IT work together** to ensure the uniformity, accuracy, stewardship, semantic consistency, and accountability of the enterprise's shared Master Data assets" (Gartner)

- **Planning for Master Data Management** includes several basic steps:
 - Identify Master Data entities
 - Develop rules for matching and merging
 - Correct mismatches
 - **Distribute trusted data**

Reference Data

It is any data used to characterize or **classify other data**, or to relate data to information external to an organization (Chisholm, 2001)

- Reference Data Structure (Lists, Taxonomies, Ontologies)
- Proprietary or Internal Reference Data (to support internal applications)
- **Industry Reference Data**
- Geographic or Geo-statistical Data
- Computational Reference Data

Reference Data is data that defines the set of **permissible values** to be used by other data fields

- **Units of measurement, country codes, corporate codes, fixed conversion rates, and calendar structure are some examples**

